CHIANG MAI UNIVERSITY

Bachelor of Science (Software Engineering)

College of Arts, Media and Technology

SE233 Advanced Programming

Term project I

1) Preface

- The first term project of SE233 every year involves having students develop some parts of an application suite, essentially covering the functionalities practiced during the first half of the course. These functionalities include GUI, events, exception handling, and multiprocessing.

  The selected application to mimic for this year is a vector graphics conversion utility, drawing inspiration from the inner interface of applications like Vector Magic. The goal is to create a desktop application that performs batch raster-to-vector image conversions (converting PNG/JPG to SVG). Completing this project will provide practical experience in handling file I/O, managing external CPU-bound processes, and building responsive user interfaces.

  You can determine the desired score by selecting features described in the scoring rubric provided at the end of this document. Additionally, the GUI does not need to be entirely identical to the provided web application example.

2) Setup

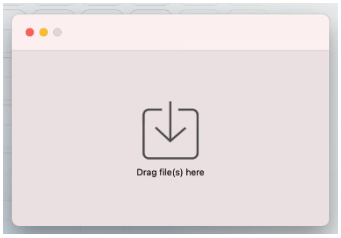
- **Work in pairs** (solo or group of three can be negotiable with a scoring handicap).
- You can form pairs across different sections.
- You will provide an **onsite demonstration** for grading, and the two lecturers will evaluate your work based on the scoring rubrics outlined in Section 7 of this document. The final score will be the average of the scores determined by the two lecturers.
- **Assignment given on:** September 7th, 2026.
- **Code submission due date:** October 13th, 2026 (approximately 5 weeks).
- **(Tentative) Onsite demonstration date:** October 14th, 2026.

3) Scores

- 15 points from your final grade.

4) Functional requirements:

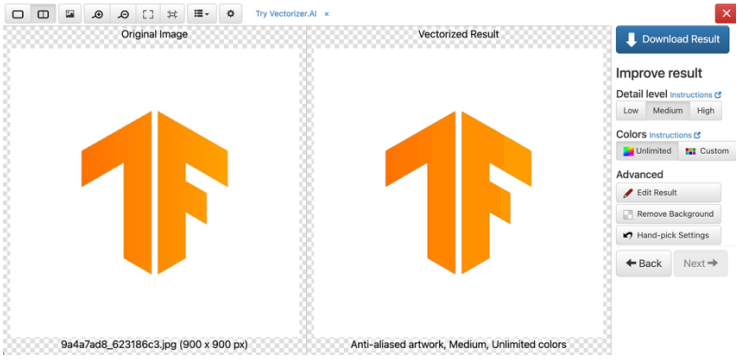
- Drop zone



The application shall support drag-and-drop functionality for file input. The permissible file types that can be dropped into this zone include:

- A single image file (.jpg, .png)
- Multiple image files in jpg and/or png format
- A single archive file in .zip format containing images in multiple format

- Main interface layout



Once images are dropped, the application will transition to the main editor interface, which is divided into three sections:

- **Center preview**: Displays the Original Image and the Vectorized Result side-by-side.
- **Top panel**: Must contain only the following three controls:
  - Zoom In
  - Zoom Out
  - Zoom to Fit
- **Right-handed panel**: Must contain the configuration settings and export controls.
  - **Export all button**: Replaces the traditional download button. Since all configurations are made dynamically, this button initiates the batch save process.
  - **Detail level**: A selector with three options: Low, Medium, and High.
  - **Colors**: A selector with two options:
  - **Unlimited**: Uses the original color palette.
  - **Custom**: Limits the vector output to a specific number of colors. The application must auto-detect the image's colors. If the image has more than 5 colors, cap the custom selection limit at 5. **This setting must be pre-selected to 2 colors by default.**
  - **Remove background**: A checkbox to output a transparent SVG.
  - **Navigation**: **Back** and **Next** buttons to flip through and preview the multiple images dropped into the application.
  - ***Edit result*** *and* ***Hand-pick settings*** *from the original Vector Magic UI are explicitly* ***excluded*** *from this project scope).*
- Batch processing & exporting
  - After clicking the **Export All** button, the application will prompt the user to specify an output folder path using the default file browser of the operating system.

- **Constraint-triggered parallel processing:** If multiple files are dropped, the application must process the images concurrently (in parallel) **ONLY IF** the settings are configured to **Medium Detail** and **Custom Colors (up to 5, defaulting to 2).**
- Upon confirmation, the application will process and save the resulting .svg file(s) to the selected output path.
- ***Additional note:*** You can integrate any available library in Github into your workflow. The lecturers recommend using a Java wrapper for the **Potrace** or **AutoTrace** command-line tools for the heavy raster-to-vector calculation.

5) **Additional specifications / non-functional requirements:**

- The application shall correctly handle .jpg and .png image formats and output valid .svg files.
- The application shall correctly extract and process images from .zip archives.
- The application shall provide a clear and intuitive user interface avoiding thread freezing during heavy vector tracing.
- The application shall handle potential runtime errors gracefully, providing informative messages to the user (e.g., missing CLI tools, corrupted images).

6) **Scoring rubrics**

- **If working in pair (group of 2 students)**

Item	To get 1 point	To get 2 points	To get 3 points
Vectorization Format	Vectorization correctly converts both jpg and png formats to SVG.	-	-
UI Navigation & View	Top panel Zoom In/Out and Zoom to Fit functions work correctly for the preview.	One point plus the Right panel Back/Next buttons correctly navigate through multiple dropped images.	-
Configuration: Details & BG	The user can change the Detail level (Low, Med, High), and it reflects in the output. The preview must be available from this configuration.	One point plus the Remove Background checkbox works and successfully outputs a transparent SVG.	-
Configuration: Colors	The Unlimited color option works correctly.	One point plus the Custom color option works, correctly detects colors, caps at a maximum of 5 colors, and defaults to 2 colors upon load.	Two points plus all these configurations have an instant preview available and work correctly.
Zip Handling	The application accepts file(s) in zip format, extracts them, and processes them as if multiple files were dropped.	-	-

Preprocessing	When multiple files are dropped, parallel execution triggers preprocessing sequentially: detect color, configure with medium detail, two colors, and pre-show the result.	One point plus these preprocessing tasks are done in parallel, and the UI clearly indicates (e.g., via a progress bar) that files are being processed concurrently without freezing.	-
Export	The export is correctly implemented and the output file can be opened from the OS.	One point plus this export process is done in parallel, and the UI clearly indicates (e.g., via a progress bar) that files are being processed concurrently without freezing.	-
Additional feature	Adds one additional feature not presented in this list. The feature must be significant enough to be justified by the two lecturers.	-	-
UX and UI	Subjective ratings are averaged by the two lecturers.	-	-

- If solo work or working in group of 3 students
  - Plus, skipping an item from the scoring rubric will **cost a 1-point penalty** for each item skipped.

Item	To get 1 point	To get 2 points	To get 3 points
Vectorization Format	Vectorization correctly converts both jpg and png formats to SVG.	-	-
UI Navigation & View	-	Top panel Zoom In/Out and Zoom to Fit functions work correctly for the preview, **and** the Right panel Back/Next buttons correctly navigate through multiple dropped images.	-
Configuration: Details & BG	-	The user can change the Detail level (Low, Med, High), and it reflects in the output. The preview must be available from this configuration, **and**	-

		the Remove Background checkbox works and successfully outputs a transparent SVG.	
Configuration: Colors	-	-	The Unlimited color option works correctly, **and** the Custom color option works, correctly detects colors, caps at a maximum of 5 colors, and defaults to 2 colors upon load, **and** all these configurations have an instant preview available and work correctly.
Zip Handling	The application accepts file(s) in zip format, extracts them, and processes them as if multiple files were dropped.	-	-
Preprocessing	-	When multiple files are dropped, parallel execution triggers preprocessing sequentially: detect color, configure with medium detail, two colors, and pre-show the result, **and** these preprocessing tasks are done in parallel, and the UI clearly indicates (e.g., via a progress bar) that files are being processed concurrently without freezing.	-
Export	-	The export is correctly implemented and the output file can be opened from the OS, **and** this export process is done in parallel, and the UI clearly indicates (e.g., via a progress bar) that files	-

		are being processed concurrently without freezing.	
Additional feature	Adds one additional feature not presented in this list. The feature must be significant enough to be justified by the two lecturers.	-	-
UX and UI	Subjective ratings are averaged by the two lecturers.	-	-